Aamir Talal

Applied ML Research | AI Engineer, Safran Aircraft Engines | M.Eng Computer Science (Machine Learning)
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Education

CY Tech — Engineering Cycle, Computer Science, Machine Learning specialisation 2026–2029
Top 5% of the cohort; machine learning, statistics, optimisation, data science *Paris, France*

CY Tech — Integrated Preparatory Cycle 2024–2026
Mathematics, physics and computer science *Paris, France*

Experience

AI Engineer, Applied Research — Safran Aircraft Engines 2026–2029
Apprenticeship *Montereau-sur-le-Jard, France*

- Design and deploy LLM agents that automate and digitalise sales administration (ADV) processes
- Build and evaluate agentic workflows that ground generative AI in existing business tools and data
- Work directly with business teams to identify, scope and prioritise automation use cases

Research Projects

Remaining Useful Life Prediction for Turbofan Engines | *Predictive maintenance, NASA C-MAPSS* 2026

- Framed predictive maintenance as RUL regression on NASA C-MAPSS FD001 (100 run-to-failure engines, 21 sensors), piecewise-linear target capped at 125 cycles
- Designed a leakage-free protocol: engine-level grouped cross-validation and causal rolling-window features (means, slopes) that never look past the current cycle
- Benchmarked a from-scratch gradient-descent linear baseline against Ridge, Random Forest and LightGBM, scored with RMSE, MAE and the asymmetric NASA PHM08 score, which penalises late predictions
- Interpreted models with SHAP and error analysis by distance to failure, linking dominant sensors to the observed degradation
- Stack: Python, pandas, scikit-learn, LightGBM, SHAP

Legible: speaker-attributed grounding for legal LLMs | *Winner, Mistral×Stanford Law School Hackathon* 2026

- Designed the System 1→System 2 engine: rule-based zoning and a calibrated classifier (Jev) attribute each sentence of a French court ruling to its speaker with a confidence score, conditioning a Mistral LLM that answers with paragraph-level citations
- On the team's 25-ruling benchmark, 4× fewer misattributed answers than a baseline LLM (11 vs 44 out of 224); live demo: legible-or4n.onrender.com
- Stack: Python, FastAPI, Mistral API, MCP

MiroPolis: multi-agent LLM simulation of parliament | *Top Project, National Assembly Hackathon* 2026

- Built a multi-agent digital twin (OASIS) in which LLM deputies, seeded with the chamber's real composition, groups and programmes, debate, amend and vote on a bill
- Compared fine-tuned agents with persona-orchestrated pre-trained LLMs; pivoted to the latter, faster and more realistic

Skills

Machine Learning: regression, tree ensembles (Random Forest, LightGBM), cross-validation, SHAP, PyTorch, Hugging Face
AI/LLM: LLM agents, RAG, prompt engineering, LLM evaluation, MCP, LangChain, OpenAI API
Data & Orchestration: SQL, pandas, NumPy, scikit-learn, data pipelines, n8n
Languages: Python, C, SQL, JavaScript, Shell/Bash, PHP
Spoken languages: French (native), English (C1), Urdu (C1)

Hackathons & Achievements

- Winner, Mistral×Stanford Law School Hackathon, and winner of the Legora track
- Top Score, IBM×Airbus Hackathon (Télécom Paris)
- Top Project, French National Assembly Hackathon
- Top 20 out of 300, Anthropic×Hugging Face Hackathon
- ML algorithms and maths implemented from scratch on Deep-ML; 50+ LeetCode problems solved

Community & Interests

- Content creator, Intellectuel Athlète (personal development / tech): 14,000+ followers, 2–3M monthly views
- Volunteering: 50+ weekly humanitarian outreach rounds (maraudes), Au cœur de la précarité; created an Arabic-learning web app (1,000+ users in 24 h)
- Sport: Brazilian jiu-jitsu and grappling (competition), running